DBMS ER Diagram Solutions

Hospital Management | Online Shopping | University Course Registration

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Diagram key (Chen notation): rectangle = entity, diamond = relationship, small oval = relationship attribute, underlined attribute = primary key, bold/thick edge = total participation, thin edge = partial participation.

1. Hospital Management System

Entities & primary keys

- **Department**(Dept_ID **PK**, Dept_Name, Location)
- **Doctor**(Doctor_ID **PK**, Name, Specialization, Qualification, Experience)
- **Patient**(Patient_ID **PK**, Name, Age, Gender, Contact_No, Address)
- **Ward**(Ward_ID **PK**, Ward_Name, Floor)
- **Room**(Room_No **PK**, Type)

Relationships & constraints

- **Employs** (Department–Doctor): 1:N, total participation on Doctor
- **Treats** (Doctor–Patient): M:N, with attributes Treat_Date, Diagnosis → becomes an associative entity
- **Admitted_in** (Patient–Room): N:1, partial participation on Patient
- **Part_of** (Room–Ward): N:1, total participation on both sides

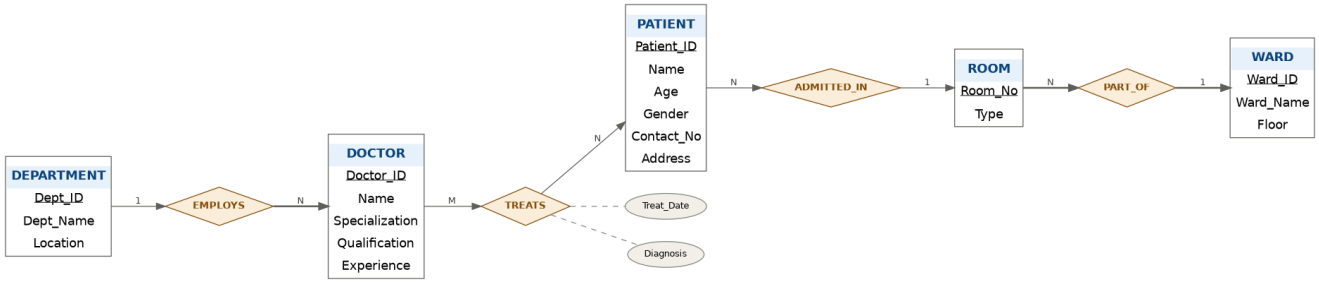


Fig. 1 – ER diagram for the Hospital Management System

Relational schema

```
Department(Dept_ID, Dept_Name, Location)
Ward(Ward_ID, Ward_Name, Floor)
Room(Room_No, Type, Ward_ID)                        -- FK Ward_ID -> Ward
Doctor(Doctor_ID, Name, Specialization, Qualification, Experience, Dept_ID)
                                                    -- FK Dept_ID -> Department
Patient(Patient_ID, Name, Age, Gender, Contact_No, Address, Room_No)
                                                    -- FK Room_No -> Room
Treats(Doctor_ID, Patient_ID, Treat_Date, Diagnosis)
                                                    -- PK (Doctor_ID, Patient_ID)
                                                    -- FK Doctor_ID -> Doctor, Patient_ID -> Patient
```

2. Online Shopping System

Entities & primary keys

- **Customer**(Customer_ID **PK**, Name, Email, Phone, Address)
- **Order**(Order_ID **PK**, Order_Date, Total_Amount)
- **Product**(Product_ID **PK**, Product_Name, Price)
- **Category**(Category_ID **PK**, Category_Name)
- **Seller**(Seller_ID **PK**, Seller_Name, Contact)
- **Payment**(Payment_ID **PK**, Amount, Payment_Mode, Payment_Status)
- **Shipping**(Shipping_ID **PK**, Address, Status, Delivery_Date)

Relationships & constraints

- **Places** (Customer–Order): 1:N, total participation on Order
- **Contains** (Order–Product): M:N, attributes Quantity, Price → associative entity Order_Item
- **Belongs_to** (Category–Product): 1:N, total participation on Product
- **Sells** (Seller–Product): M:N, attribute Stock_Qty → associative entity Sells
- **Has** (Order–Payment): 1:1, total participation on both sides
- **Has** (Order–Shipping): 1:1, total participation on both sides

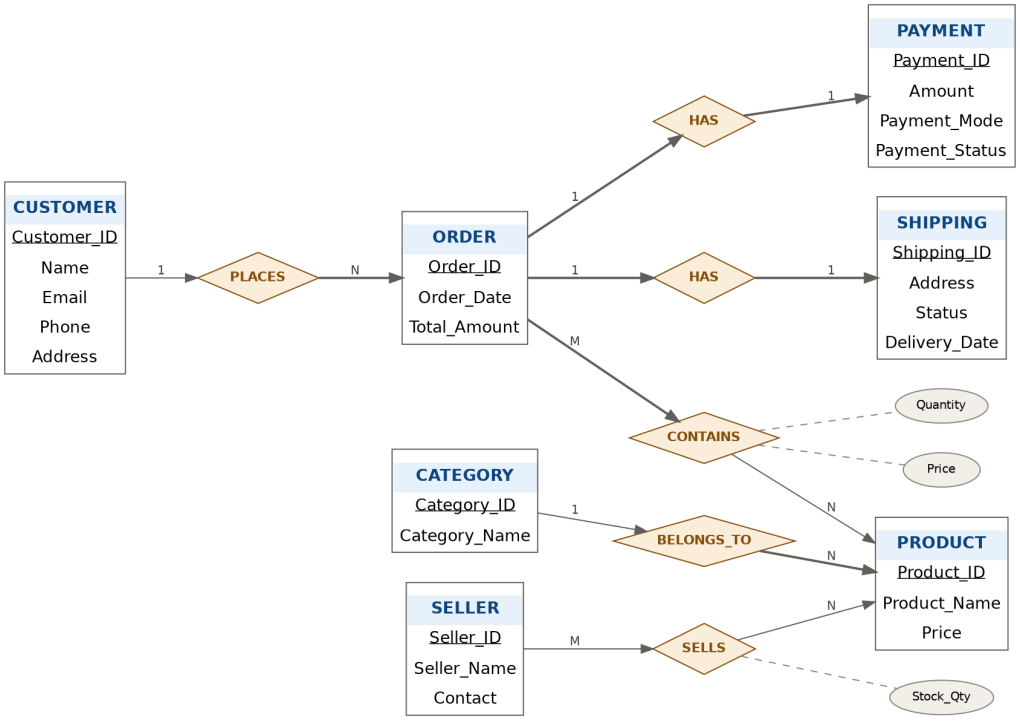


Fig. 2 – ER diagram for the Online Shopping System

Relational schema

```
Customer(Customer_ID, Name, Email, Phone, Address)
Category(Category_ID, Category_Name)
Product(Product_ID, Product_Name, Price, Category_ID)     -- FK Category_ID -> Category
Seller(Seller_ID, Seller_Name, Contact)
Order(Order_ID, Order_Date, Total_Amount, Customer_ID)    -- FK Customer_ID -> Customer
Order_Item(Order_ID, Product_ID, Quantity, Price)         -- PK (Order_ID, Product_ID)
                                                            -- FK Order_ID -> Order, Product_ID -> Produc
Sells(Seller_ID, Product_ID, Stock_Qty, Price)             -- PK (Seller_ID, Product_ID)
```

-- FK Seller_ID -> Seller, Product_ID -> Prod

Payment(Payment_ID, Order_ID, Amount, Payment_Mode, Payment_Status)

-- FK Order_ID -> Order (unique)

Shipping(Shipping_ID, Order_ID, Address, Status, Delivery_Date)

-- FK Order_ID -> Order (unique)

3. University Course Registration System

Entities & primary keys

- **Department**(Dept_ID **PK**, Dept_Name)
- **Course**(Course_ID **PK**, Title, Credits, Semester_Offered)
- **Student**(Student_ID **PK**, Name, Email, Semester)
- **Professor**(Professor_ID **PK**, Name, Email, Designation)

Relationships & constraints

- **Offers** (Department–Course): 1:N, total participation on Course
- **Enrolls** (Student–Course): M:N, attributes Semester, Grade → associative entity Enrollment
- **Teaches** (Professor–Course): M:N, attribute Semester → associative entity Teaches

Enrollment and Teaches are the two associative (bridge) entities required to resolve the M:N relationships; Enrollment additionally stores the Grade attribute, which can only be modeled as an attribute of the enrollment record, not of Student or Course alone.

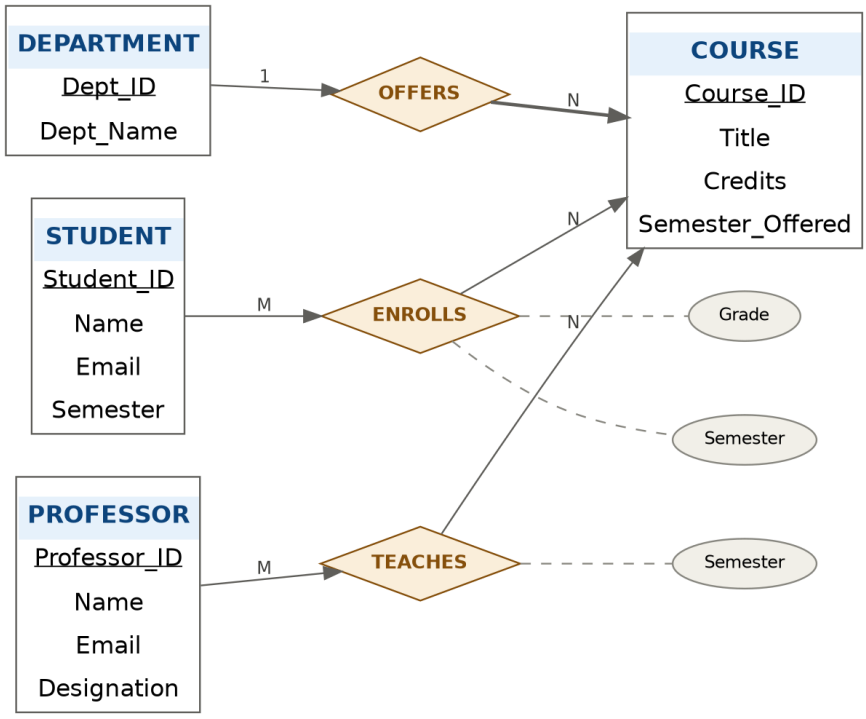


Fig. 3 – ER diagram for the University Course Registration System

Relational schema (in 3NF / BCNF)

```
Department(Dept_ID, Dept_Name)
Course(Course_ID, Title, Credits, Semester_Offered, Dept_ID)   -- FK Dept_ID -> Department
Student(Student_ID, Name, Email, Semester)
Professor(Professor_ID, Name, Email, Designation)
Enrollment(Student_ID, Course_ID, Semester, Grade)   -- PK (Student_ID, Course_ID, Semester)
                                                       -- FK Student_ID -> Student, Course_ID -> Course
Teaches(Professor_ID, Course_ID, Semester)            -- PK (Professor_ID, Course_ID, Semester)
                                                       -- FK Professor_ID -> Professor, Course_ID -> Cour
```